



The Secret Code Breakers

An Introduction to Regular Expressions (Regex)

Grade 6 | Session 1

Planrs Academy Content Production

The Library Mystery

Imagine you are in a big library. You find a book on a shelf, but its title is partly torn. You can only see some of the letters.

Title: T I L _

Could it be **TILE**? Or maybe **TILL**? Or **TILT**?

In the world of computers, we often have "missing pieces" of information. **Secret Codes** (officially called **Regular Expressions** or **Regex**) help us find things even when we don't know the exact word!

What is Regex?

Regex is like a secret language that you use to talk to a computer. Instead of telling the computer to look for one exact thing, you give it a **pattern**.

Normal Search

Looking for a specific house address: **House No. 123**.

Regex Search

Looking for **ANY** house that starts with the number 1 and has 3 digits total.

It's a superpower for **Data Detectives**! It helps us sort through millions of emails, phone numbers, and codes in seconds.

Meet the Dot (.)

The **Dot** is the "Wildcard" of the Regex world. It represents **any single character** (a letter, a number, or even a space).

Example: The "R.m" Mystery

If you search for the pattern:

```
R . m
```

The computer will find matches like:

- **Ram** (The dot becomes 'a')
- **Rim** (The dot becomes 'i')
- **Rom** (The dot becomes 'o')

It's like a puzzle piece that fits anywhere!

The Plus (+) Sign

The **Plus** sign means "one or more of the previous character." It tells the computer that a certain letter or symbol can repeat itself!

Example: The Cheering Crowd

If you search for the pattern:

```
go +
```

The computer will find:

- **go** (one 'o')
- **goo** (two 'o's')
- **gooo** (three 'o's!')

This is useful for finding where people might have gotten excited and typed extra letters in a message.

The Asterisk (*)

The **Asterisk** is similar to the Plus, but with a twist: it means **zero or more** of the previous character. This means the letter could be there many times, or not at all!

Example: The Spelling Helper

In different parts of the world, words are spelled differently. Take the word "color" vs "colour".

```
colou*r
```

This pattern matches:

- **color** (zero 'u's)
- **colour** (one 'u')

It's great for finding words with optional letters!

Character Classes []

Sometimes you don't want "any" character. You want a specific group, like only numbers or only certain letters. We use **Square Brackets** for this.

[0-9] means any single digit from 0 to 9.

[a-z] means any lowercase letter from a to z.

Indian Context: Mobile Numbers

Indian mobile numbers often start with **+91**. A computer can use a pattern to check if a phone number is typed correctly by looking for 10 digits in a row using these classes!

Guided Practice

Activity: The Postman's Helper

In India, every area has a 6-digit **PIN Code**. These codes always start with a number from 1-9 and then have 5 more digits (0-9).

The Regex Pattern:

```
[1-9][0-9]{5}
```

Let's test it on this text:

"Send the gift to 400001 Mumbai, then to 110001 Delhi. Don't send it to 12345 Bangalore because that code is too short!"

✓ **400001** - Matched!

✓ **110001** - Matched!

✗ **12345** - Not matched (only 5 digits).

Pause & Think #1

Computers deal with billions of pieces of data every day. Think about how many emails, phone numbers, and Aadhaar numbers exist in India!

Discussion Question:

Why do you think computers need patterns like Regex instead of just humans checking every single number manually?

Share your thoughts with your classmates. Remember, there are no wrong answers—just curious minds exploring!

Independent Challenge

Task: Find "India" Even if Misspelled!

Sometimes people make small typos. They might spell our country as "**India**" or "**Indya**".

Your goal is to create a single Regex pattern that finds both versions.

Hint: Use the **Dot (.)** to replace the one letter that changes between the two spellings.

Your Solution:

Ind _ a

Test Your Secret Code Skills!

1. What does the pattern "R.m" match?

- a) Only "Ram"
- b) "Ram", "Rim", and "Rom"
- c) "Room"

2. Which symbol means "one or more" of the letter before it?

- a) Dot (.)
- b) Asterisk (*)
- c) Plus (+)

3. How many digits are in an Indian PIN code?

- a) 5 digits
- b) 6 digits
- c) 10 digits

Answers: 1: b, 2: c, 3: b

Detective's Cheat Sheet

Keep this guide in your pocket for your next data mystery!

. (Dot)

Any single character.

+ (Plus)

One or more times.

*** (Asterisk)**

Zero or more times.

[0-9]

Any single digit.

Stay curious and keep practicing your patterns!